FreeBuff v4

Multi-Modal Brain Tumor Classification & Clinical Decision Support

Diagnostic Imaging Classification Report

Generated By	FreeBuff v4 Research System
Report Date/Time	2026-09-06 16:36:36
Modality	MRI Brain
Scan Description	Brain MRI slice — classification target: Glioma
Patient ID	TEST_PT_001
Patient Name	John Doe (de-identified)
Exam Date	2026-09-06

CLASSIFICATION RESULTS

Predicted Class: Glioma

Prediction Confidence: 0.9123 (91.23%) — Confidence Band: High

Predicted Class Index: 0

Test-Time Augmentation (TTA) Used: True

Source Image: test_mri_001.dcm

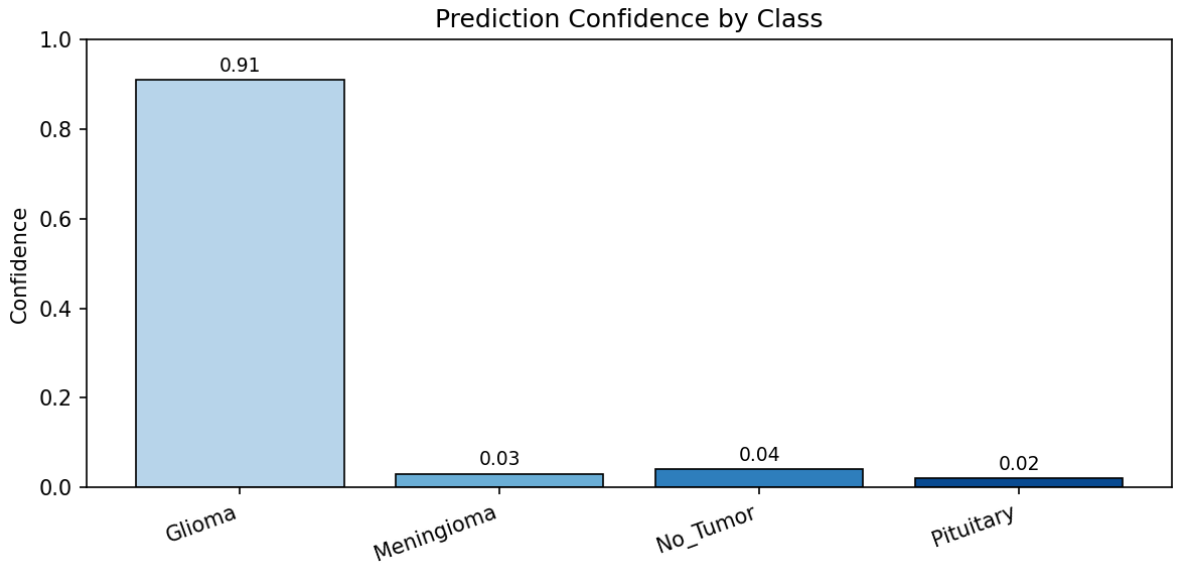
Per-Class Probability Distribution

Class	Probability	Percentage
Glioma	0.9100	91.00%
Meningioma	0.0300	3.00%
No_Tumor	0.0400	4.00%
Pituitary	0.0200	2.00%

Raw Logits (Pre-Softmax)

Class	Logit
Glioma	2.1000
Meningioma	-0.5000
No_Tumor	0.0000
Pituitary	-0.8000

CONFIDENCE SPECTRUM VISUALIZATION



The bar chart above displays the predicted probability (confidence) assigned to each tumour class by the ensemble decision pathway. Higher bars indicate stronger model assertion for the corresponding diagnostic category.

GRAD-CAM++ EXPLAINABILITY HEATMAP

[No Grad-CAM++ overlay available for this case.]

The heatmap above visualizes the image regions most influential in the model's classification decision (Grad-CAM++). Warmer colors indicate higher importance. This visualization is intended to assist clinician review and is NOT a standalone diagnostic indicator.

WHO GRADE & STAGING ASSESSMENT

Tumour Class: **Glioma**

Grade Label: N/A

Typical Imaging Features: N/A

Malignancy Profile: N/A

Growth Pattern: N/A

Recommended Intervention Protocol: N/A

CLINICAL REASONING NARRATIVE

Predicted class: Glioma Confidence: 91.2%

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